

SEMESTER III

Course Name: Web Technologies & E-Commerce	
Semester 3	
Course Code:	BCA-DC301
Course Title:	Web Technologies & E- Commerce
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Prerequisites	NIL
Course Category:	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome

After studying this course, students will be able to:

In this course, the learners will be able to develop expertise related to the following:

- Students should be able to design and implement a basic website.
- Students should be able to implement different navigation strategies.
- Students should be able to develop simple back-end database to support a website.
- Students should be able to recognize and evaluate website organizational structure and design elements.

Unit		Hours
1	HTML BASICS AND SERVERS	12
1.1	History of the Internet and World Wide Web, Search Engines, News-group, E-mail and its Protocols, Web Portal, Browsers and their versions, Its functions, URLs, web sites, Domain names, Portals. Introduction to web applications, HTML, Client Side Scripting Vs Server Side Scripting,	
1.2	Web Servers : Local Servers and Remote Servers, Installing Web servers, Internet Information Server (IIS) and Personal Web Server (PWS)	
1.3	Static Web Development: HTML - Introduction to HTML, HTML Document structure tags, HTML comments, Text formatting, inserting special characters, anchor tag, adding images and Sound, lists types of lists, tables, frames and floating frames, Developing Forms, Image maps. Static website vs Dynamic website development. Cascading Style Sheet, Introduction to JavaScript.	
2	PHP-ARRAYS AND FILE	10
2.1	Introduction to PHP, Start and End Tags of PHP, Data types in PHP, Variables, Constants, operators and Expressions, printing data on PHP page, Control statements – if, switch case, for, while, do while.	
2.2	Arrays: Initialization of an array, Iterating through an array, Sorting arrays, Array Functions, Functions: Defining and Calling Functions, Passing by Value and passing By references, Inbuilt Functions.	
3	FORMS AND SESSION	10
3.1	Working with Forms: Get and Post Methods, Query strings, HTML form controls and PHP	
3.2	Maintaining User State: Cookies, Sessions, and Application State.	
3.3	Working With Files: Opening and Closing Files, Reading and Writing to Files, Getting Information on Files	
4	DATABASE CONNECTIVITY	8
4.1	PHP Database Connectivity: Introduction to MYSQL, Creating database and other operations on database, connecting to a database, Use a particular database, Sending query to database.	
4.2	Parsing of the query results, Checking data errors.	

List of Practical

Note: Practical may be covered as per the syllabus by the faculty.

Learning Approach

To understand HTML, PHP and Database connectivity with PHP.

References/suggested learning resources

Books

Textbooks:

- [T1] The complete reference HTML, by Thomas A powell, TMH publication.
- [T2] Mastering HTML 4.0 by Deborah S. Ray and Erich J. Ray. BPB Publication.
- [T3] Programming PHP. Rasmus Lerdorf, Kevin Tatroe. (O'Reilly, ISBN 1565926102).
- [T4] PHP, MySQL, and JavaScript: A Step-By-Step Guide to Creating Dynamic Websites Robin Nixon O'Reilly Media; 1 edition

Reference books:

- [R1] Core PHP Programming. Leon Atkinson (Prentice Hall, ISBN 0130463469).
- [R2] Beginning PHP5 and MySQL: From Novice to Professional, W. Jason Gilmore, 200 Apress, ISBN: 1-893115-51-8

Course Name: Computer Graphics	
Semester 3	
Course Code:	BCA- DC302
Course Title:	Computer Graphics
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Prerequisites	Basic Knowledge of Mathematics Concept & Programming Language Skill
Course Category:	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome

After studying this course, students will be able to:

In this course, the learners will be able to develop expertise related to the following: -

- Basic building blocks and core concepts of computer graphics.
- To understand Basic Graphics Primitives.
- Theoretical, mathematical foundation, and practical aspects of different graphics algorithms.
- Fundamental concepts of animation and its related technologies to design interactive graphics applications.

unit	Title	Hou rs
1	Introduction of Graphics	10
1.1	Introduction: Computer Graphics and its Applications, Overview of Graphics Systems–Video display devices, Raster-Scan Systems, Random-Scan Systems, Graphics Software.	
1.2	Scan Conversion: Scan conversion of Point and Line, Line Drawing Algorithms: DDA, Bresenham's, Circle: Mid-Point, Bresenham's,	
1.3	Animation: Introduction to Animation, Principles, Animation Methods, Animation Tools	
2	Transformation	8
2.1	Transformations: Two-Dimensional and Three-Dimensional Geometric Transformations: Translation, Scaling, Rotation, Reflection, Shearing, Homogeneous Coordinates, Composite Transformations, Rotation about an arbitrary point, Reflection about an Arbitrary Line, Affine Transformations	
2.2	Window to View port: Windowing, 2D-Viewing Pipeline, Computing Location of Viewport, Window to Viewport Transformation	
3	Clipping , Projection and Solid Modelling	12
3.1	Clipping: Introduction, Point Clipping, Line Clipping, Line Clipping Algorithms: Cohen Sutherland Algorithm, Polygon Clipping, Polygon Clipping Algorithms: Sutherland Hodgeman Polygon Clipping	
3.2	Projections: Introduction to Projections, Types of Projections: Parallel and Perspective Projections, Parallel Projections: Oblique, Orthographic, Axonometric, Perspective Projections: One-Point, Two-Point, Three-Point, Vanishing Points, Perspective Anomalies, Comparison of Parallel and Perspective Projections	
3.3	Solid Modelling: Solid Representation, Sweep Representation, Boundary Representation (B-Rep), Octrees, Constructive Solid Geometry (CSG), Comparison of Representations	

4	Curves and Surfaces, Hidden Surface Removal	10
----------	--	-----------

4.1	Curves and Surfaces: Curves Representation, Parametric and Non-Parametric Curves, Bezier Curves: Cubic and Higher-Order Curves, Blending Function, Bernstein Polynomial, Continuity Conditions, Limitations, B-Splines: Construction of B-Spline Curves, Cubic B-Spline Curves, Knot Vectors, Uniform, Open- Uniform, Non-Uniform, Non-Uniform Rational Basis Splines (NURBS), Introduction to Bezier and B-Spline Surfaces.	
4.2	Hidden Surface Removal: Need for Hidden Surface Removal, Categories of Hidden Surface Removal Methods, Z-buffer, and A-buffer, Depth Sorting, BSP-Tree, Back-Face Removal, Scan- Line Method, Area-Subdivision Method.	

List of Practical

Practical Based on Syllabus (May be Implemented in C/ C++ Programming Language)

- WAP to Draw Line Using DDA Algorithm
- WAP to Draw Line Using Brasenham's Algorithm
- WAP to Draw Circle Using DDA Algorithm
- WAP to Draw Circle Using Brasenham's Algorithm
- WAP to Translate an Object
- WAP to Rotate an Object about 60 degrees in clockwise direction
- WAP to Scale an Object to twice its size
- WAP to Clip an Object

Note: Additional Practical may be covered as per the syllabus by the faculty

Learning Approach

In this course, the learners will be able to develop expertise related to the following -

1. Demonstrate graphics drawings with respect to Graphics Primitives.
2. Apply 2D & 3D transformation concepts to represent images with different dimensions
3. Analyse and evaluate the concepts of projection and shading methods to obtain scenes with different clipping
4. To Understand Fundamental concepts of animation and its related technologies to design interactive graphics applications.

References/suggested learning resources

Books

Textbooks:

1. Donald Hearn and Pauline Baker, "Computer Graphics with C Version ", Pearson, 2ndEdition,2002.
2. Foley James D, "Computer Graphics: Principles and Practice", Addison-Wesley Professional,3rdEdition,2013.
3. Rogers and Adams, "Mathematical Elements for Computer Graphics", McGraw-Hill, 2ndEdition, 2002.

Reference books:

1. John M. Blain, "The Complete Guide to Blender Graphics: Computer Modelling & Animation", CRC Press, 5thEdition,2019.
2. Malay K. Pakhira, "Computer Graphics Multimedia Animation", PHI,2ndEdition,2010.
3. F. S. Hill, Jr. Stephen M .Kelley, "Computer Graphics using OpenGL", Pearson,3rdEdition,2008.
4. Rajesh K Maurya, "Computer Graphics with Virtual Reality System", Wiley, 2ndEdition, 2014.
5. Rogers, "Procedural Element of Computer Graphics", McGraw- Hill,2ndEdition,2001.
6. Zhigang Xiang and Roy Plastock, "Computer Graphics Schaum's Outlines Series", McGraw-Hill,2ndEdition,2015.

Course Name: Python Programming	
Semester 3	
Course Code	BCA-DC303
Course Title	Python Programming
Number of Credits	Total: (L: 2 T:1 P:2)
Total Number of teaching hours	40 Hours
Pre-requisites	NIL
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome

- Design a solution to the problem using an Algorithm, Flowchart, and Pseudocode.
- Use python programming constructs, arrays, strings, functions, and modules to solve real-world problems.
- Design & develop custom functions & modules.
- Apply data structure primitives like lists, tuples, sets, and dictionaries.
- Apply file management operations to operate on files.
- Design and develop object-oriented solutions to problems using python programming language.
- Develop GUI & networking applications using python.
- Develop python applications with database connectivity operations.
- Familiarize and apply basic data science operations using Numpy, Pandas and Matplotlib library.

Detailed Contents

Unit	Title	Hou rs
Unit 1	Principles of Programming Language and Python Preliminaries	10
1.1	Concepts of Programming Language: Introduction; Programming Domains, Language Evaluation Criteria, Influences on language design, Programming Paradigms, Implementation Methods: Compilation, Pure Interpretation, and Hybrid Implementation System.	
1.2	Revisit to Program Design: Algorithm, Flowchart & Pseudocode: Steps involved, Detailed Construction, Problems on Selection, Looping and Flow chart analysis.	

1.3	Introduction to Python: Features of Python, Structure of Python Program, Interactive Execution, Installation and working on Python IDE: Jupyter/PyCharm/IntelliJ IDEA and others, Identifiers, Keywords, Delimiters and Literals, Statements, Variables, Escape Sequences; Comments, Data Types, Type Conversion Functions and Rounding, Operators: Arithmetic, Assignment, Unary Minus, Relational, Logical, Boolean, Bitwise, Membership: in, not in, Identity: is, is not; Expressions, Operator Precedence and Associativity, Input and Output Statements, Command Line Arguments, Short-circuit, and Lazy Evaluation.	
1.4	Control Flow Statements: Conditionals: if Statement, if-else Statement, Conditional Expressions; Nested Conditionals: Nested if and Multi-Way if-elif-else Statements; Looping: while statement, for statement, else suite, The range() Function; Nested loops; The break statement; The continue Statement; The pass Statement; The assert statement; The return Statement.	
Unit 2	Functions, Modules and Built-in Primitive Data Structures in Python	10
2.1	Functions: Built-in, User-Defined and Anonymous function; Elements of User-Defined Functions, Arguments and Return Values, Formal vs. Actual Arguments, Scope and Lifetime, Positional, Keyword Arguments & Default Arguments, Nested Functions, Using Lambdas with filter(), map() and reduce() function, Decorators, Iterators, Generators, Recursion.	
2.2	Modules: Importing Modules; Math, Random and other standard library modules, Packages, Custom Modules.	
2.3	Built-in Primitive Data Structures Arrays: Single dimensional, Multi-dimensional arrays (up to three dimensions), Array Creation using array, linspace, logspace, arrange, zeros, ones, Operations on Arrays. Strings: Basics, Immutability, String creation, String Indexing and Slicing, String Manipulation, The subscript operator, Searching substrings.	
Unit 3	File Management and Object Oriented Programming	10

3.1	File Handling: Text and Binary Files - Writing and Reading Operations, Random Access to Files, The with Statement, Pickle in Python, Manipulating Files and Directories, Closing Files.	
3.2	Introduction to Object-Oriented Programming: Features, Classes & Objects, Immutable vs. Mutable Objects, Access Modifiers, Attributes and methods, Data Hiding, The 'self' variable, Constructor, Instance Variables and Class or Static Variables, Inner Classes, Passing members of one Class to another Class	
3.3	Exception Handling: Error, Exception: Preliminaries and Exception Class Hierarchy, Handling Exception using try, except and finally clauses, Raising Exceptions, Assertions, User-Defined Exceptions, Exception logging.	
Unit 4	Advanced Python	10
4.1	GUI Programming with Tkinter: Creating User-interface; GUI Widgets, Creating Layouts, Check Box, Radio Buttons, List Box, Menus, Menus Options, Dialog Boxes, Tables.	
4.2	Network Programming: Basics of Sockets, Socket Methods; TCP and UDP Sockets, Two-way Client-Server Communication, Sending email.	
4.3	Database Access: Advantages of a DBMS over Files, Database Connectivity Operations: Create, Insert, Select, Delete, Drop, Update & Joins.	

List of Practical

1. Exercise to set up Python Environment and IDE - Jupyter Notebook/ PyCharm/ IntelliJ IDEA/ other.
2. Exercises on variables and assigning values, type conversion.
3. Exercises on input, output and formatted output.
4. Exercises on Python operators, Escape Sequences and Comments.
5. Exercises on Operator Precedence and Associativity.
6. Exercises on number system conversion.
7. Exercises on expression evaluation.
8. Exercises on command line arguments.
9. Exercises on if statement, if...else statement, if...elif...else statement, conditional expressions, Nested Conditionals.
10. Exercises on looping, range() function, nested loops, the else suite.
11. Exercises on break statement, continue statement, pass statement, assert statement & return Statement.

12. Exercises on built-in functions.
13. Exercises on user-defined functions, passing & returning values, variable-length arguments.
14. Exercises on lambda functions, using lambdas with map(), reduce(), and filter function.
15. Exercises on Function Decorators, Generators & Iterators.
16. Exercises on nested functions & recursion.
17. Exercises on Importing python modules, using defined functions, and creating custom modules.
18. Exercises on array creation, accessing the elements & processing/manipulating them (single-dimensional & multidimensional up to three dimensions).
19. Exercises on string creation, string operations such as determination of length, indexing & slicing, the subscript operator, repeating the strings, concatenation, checking for substring & obtaining the position, comparing the strings, removing spaces, replacing substring, splitting, joining, checking starting & ending of a string, string & character testing methods, formatting strings, searching, sorting, insert substring into string.
20. Exercises on List creation and manipulation, List methods, List Comprehension, List Cloning, Searching and Sorting Lists, Tuple creation, accessing elements & operations, List to Tuple conversion & vice-versa.

Note: Additional Practical may be covered as per the syllabus by the faculty

Learning Approach

1. Design solution to the problem.
2. Master the fundamentals of writing Python scripts.
3. Use control constructs, functions & modules.
4. Develop custom functions & modules.
5. Use data structures in python.
6. Apply file handling operations and object-oriented programming techniques.
7. Explore GUI programming, network programming, and database operations in python.
8. Apply basic data science operations.

References/suggested learning resources

(a) Books

1. Concepts Of Programming Languages, Sebesta, Pearson Addison Wesley.
2. Programming languages: Design and implementation, Terrance W. Pratt, Marvin V. Zelkowitz, T.V. Gopal
3. The Art of Programming Through Flowcharts & Algorithms, Anil Bikas Chaudhuri.
4. Beginning Python From Novice to Professional, Magnus Lie Hetland, Apress
5. Python for Data Analysis, Wes McKinney, O'Reilly Media, Incorporated
6. Python Data Analytics With Pandas, NumPy, and Matplotlib, Fabio Nelli, Apress
7. Exploring Python, Timothy A. Budd, McGraw-Hill Education
8. Learning Python, Mark Lutz, O'Reilly Media, Inc.
9. Introduction to Programming Using Python, Liang Y. Daniel, Pearson Education

10. Programming in Python 3: A Complete Introduction to the Python Language, Mark Summerfield (Author), Pearson Addison-Wesley Professional.
11. Core Python Programming, R. Nageswara Rao, Dreamtech Press

Course Name: Database Management System	
Semester 3	
Course Code	BCA-DC304
Course Title	Database Management System
Number of Credits	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	1. Elementary Maths (Sets, Relations) 2. Computer fundamentals related to memory organization such as primary memory, secondary memory, etc, 3. Knowledge of basic data structures. 4. Basic knowledge of data storage and file management system
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome

- The concepts related to database, database techniques, SQL and database operations are introduced in this subject. This creates a strong foundation for application data design.
- Explain the various database components, models, DBMS architecture and Database Security
- Construct advanced SQL queries on data and apply Procedural abilities through PL/SQL.
- Examine the use of normalization and functional dependency for database design.
- Appraise the concepts of transaction, concurrency control and recovery in databases.

Detailed Contents

Unit	Title	Hours
Unit 1	Introductory Concepts of DBMS	10
1.1	Basic concepts: Database & database users, characteristics of the databases, database systems, concepts and architecture, Data Models, Schemas & instances, DBMS architecture & data independence, Overview of hierarchical, Network & Relational Database Management Systems.	4
1.2	Data Modelling using the Entity-Relationship Model: ER model concepts, notation for ER diagram, mapping constraints, Concepts of keys, Extended ER model - Generalization, Specialization, Aggregation, ER diagram to tables Mapping.	6
UNIT 2	Data Models and SQL	10
2.1	Data Models: The importance of data models, Basic building blocks, Business rules, The evolution of data models, Degrees of data abstraction, Logical view of data, keys, integrity rules.	4
2.2	Relational Model: Relational data model, Relational integrity constraints: Entity Integrity, Referential integrity, Domain Constraints, Key constraints. Relational Algebra Introduction on SQL: SQL commands and types, DML, DDL, DCL, TCL. SQL Data Types and literals, Operators in SQL. Database Objects: Table, View, Sequence, index, Synonym, Queries	6
UNIT 3	Normal Forms and PL/SQL	10
3.1	Normal Forms: Functional Dependencies, Non-loss Decomposition, First, Second, Third Normal Forms, Dependency Preservation, Boyce/Codd Normal Form, Multi-valued Dependencies and Fourth Normal Form, Join Dependencies and Fifth Normal Form.	5
3.2	PL/SQL Programming: Introduction to PL/SQL, Structure of PL/SQL Block, PL/SQL language: Operators, Control Structure, Cursors, Triggers, Procedures and functions.	5

UNIT 4	Transaction Management	10
	Transaction Management: ACID properties, serializability of Transaction, Testing for Serializability and concurrency control, Lock- based concurrency control (2PL, Deadlocks), Time stamping methods, Database recovery management, Recovery Techniques.	10

List of Practical

1. Design a Database and create required tables. For e.g. Bank, College Database
2. Apply the constraints like Primary Key, Foreign key, NOT NULL to the tables.
3. Write a SQL statement for implementing ALTER, UPDATE and DELETE
4. Write the queries to implement the joins
5. Write the queries for implementing the following functions: MAX, MIN, AVG, COUNT
6. Write the queries to implement the concept of Integrity constraints
7. Write the queries to create the views
8. Perform the queries for triggers
9. Perform the following operation for demonstrating the insertion, updation, and deletion using the referential integrity constraints

Note: Additional Practical may be covered as per the syllabus by the faculty

Learning Approach

In this course, the learners will be able to develop expertise related to the following:-

- Develop a broad understanding of database concepts and Database Management System Software, data models, schemas and instances, data constraints, relational algebra, and calculus.
- Acquire Knowledge to model an application's data requirements using conceptual modelling tools like ER diagrams and design database schemas based on the conceptual model.
- Be able to write SQL and PL/SQL commands to create and manipulate database objects.
- Be able to discuss the importance of normalization and improve the database design by applying various normal forms.
- Get in-depth knowledge of concurrency control mechanisms, transaction management techniques, and database security.

References/suggested learning resources

(a) Books

Text Books:

1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, —Database System ConceptsII, 5th Edition, Tata McGraw Hill, 2006
2. Elmasri and Navathe, —Fundamentals of Database SystemsII, 6th Ed., Pearson, 2013

3. C.J.Date, A.Kannan, S.Swamynathan, —An Introduction to Database Systemsll, 8th Edition, Pearson Education, 2006.
4. J. D. Ullman, —Principles of Database Systemsll, 2nd Ed, Galgotia Publications, 1999.
5. Vipin C. Desai, —An Introduction to Database Systemsll, West Publishing Co.

(b) Open source software and website address:

1. Oracle 10g
2. MySql

Course Name: Computer Networks	
Semester III	
Course Code	BCA-DC305
Course Title	Computer Networks
Number of Credits	Total: (L: 2 T:1 P:2)
Total Number of teaching hours	40 Hours
Pre-requisites	Basic Fundamentals of Computers
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome

- Understand basics, topologies, and working mechanisms of wired and wireless computer networks.
- Analyze the features and operations of protocols of OSI reference model & TCP/IP protocol suite.
- Design, calculate, and apply routing mechanisms for IPv4 & IPv6
- Identify the networking requirements for an organization and select & propose appropriate architecture and technologies.
- Work on Network addressing, design and implementation.

Detailed Contents

Unit	Title	Hours
Unit 1	Basics of Computer Network	10
1.1	Goals and Applications of Networks, Network structure and architecture, The OSI reference model, services, networks topology. Introduction to 5 Layer TCP/IP Network Model, Comparison with 7 Layer OSI Reference Model.	5
1.2	Physical Layer: The Physical Layer, Theoretical Basis for Data Communication, Guided Transmission Media, Wireless Transmission, Communication Satellites, Need for Modulation, Digital Signal Encoding Formats – NRZ-L, NRZI, bipolar-AMI, Manchester, Differential Manchester, Digital Modulation.	5
Unit 2	Data Link Layer	10
2.1	The Data Link Layer: Data Link Layer Design Issues, Error Detection and Correlation, Flow Control Protocols, Stop-and-wait Flow Control, Sliding – Window Flow Control, Error Control, Stop-and-wait ARQ, Go-back-N, Selective-repeat, Example of Data Link Protocols-HDLC.	5
2.2	Medium access sublayer: Channel allocations, ALOHA Protocols, Carrier Sense Multiple Access Protocols, CSMA with Collision Detection, Collision free protocols, Ethernet	5
Unit 3	Network Layer	10
3.1	Network Layer: Point-to-Point network, routing algorithms, congestion control, internetworking, Quality Control	5
3.2	The Network Layer in the Internet, IP packet, IP addresses, IPv4 Addressing, Subnetting & Supernetting, Need for NAT, Multicasting, Introduction to IPv6 Addressing, Need for IPv6 Addressing, Global, Local and Site Local Addressing, Multicast, Broadcast, Anycast and Unicast Addressing	5
Unit 4	Transport & Application Layer	10
4 : 1	Transport Layer: Design Issue, connection management, TCP window management, User Datagram Protocol, Transmission	5

	Control Protocol, Performance Issues, QOS, Transport Layer Security, IP Security, IPSec	
4 2	Application Layer: DNS, Electronic Mail, WWW, Streaming Protocols, DHCP & DHCPv6, SMTP, Telnet, SSH, POP, POP3, IMAP Network Security: Need for Network Security, Cryptography and Compression Techniques, Firewall, Various types of Cyber Threats	5

List Of Practicals

1. Study of Network devices in detail
2. Connect the computers in Local Area Network
3. Implementation of Data Link Framing method - Character Count.
4. Implementation of Data link framing method - Bit stuffing and Destuffing.
5. Implementation of Error detection method - even and odd parity.
6. Implementation of Error detection method - CRC Polynomials.
7. Implementation of Data Link protocols - Unrestricted simplex protocol
8. Implementation of data link protocols - Stop and Wait protocol
9. Implementation of routing algorithms - Dijkstra's algorithm
10. Study of Network IP Addressing

Note: Additional Practical may be covered as per the syllabus by the faculty

Learning Approach

1. Understand basics, topologies and working mechanism of wired and wireless computer networks.
2. Analyze the features and operations of protocols of OSI reference model & TCP/IP protocol suite.
3. Design, calculate, and apply routing mechanisms
4. for IPv4 & IPv6
5. Identify the networking requirements for an organization and select & propose appropriate architecture and technologies.
6. Work on Network addressing, design and implementation

References/suggested learning resources

(a) Books

1. Forouzan, "Data Communication and Networking", TMH, 4th Edition.
2. Behrouz A. Forouzan, "TCP/IP Protocol Suite", Tata Mc Graw Hill, 4th Edition, 2009.
3. A.S. Tanenbaum, "Computer Networks", PHI, 4th Edition.
4. Silvia Hagen, IPv6 Essentials. O'Reilly Media, Inc., 2006
5. W. Stallings, "Data and Computer Communication", Macmillan Press.

6. Comer, "Computer Networks and Internet", PHI. 5. Comer, "Internetworking with TCP/IP", PHI
7. Kevin R Fall and W. Richard Stevens, "TCP/IP illustrated, Volume 1: The Protocols", Addison Wesley, 2nd Edition, 2011.

(b) Open source software and website address:

1. https://onlinecourses.nptel.ac.in/noc21_cs18/preview
2. <http://www.tcpipguide.com/>
3. <https://www.coursera.org/learn/managing-network-cybersecurity/home/welcome>
4. <https://www.coursera.org/learn/wireless-communications/home/welcome>
5. <https://www.coursera.org/learn/tcpip/home/welcome>
6. <https://futureskillsprime.edcast.com/pathways/introduction-to-cyber-security>